

TOWHIDUL ISLAM

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RESEARCH INTERESTS

Artificial Intelligence for Healthcare, Medical Imaging & Image Analysis, Geometric Deep Learning, Point Cloud Learning, Computer Vision, Explainable AI, Biomedical Informatics.

EDUCATION

King Fahd University of Petroleum & Minerals (KFUPM), Dhahran, Saudi Arabia Aug 2024 – Nov 2026
(expected) (KFUPM QS Ranking 63)

M.S. in Computer Science; CGPA: **3.82/4.0**

Thesis: “A Novel Geometric Deep Learning Framework for Alzheimer’s Disease Classification Using Anatomically-Informed 2D Point Clouds from Structural MRI.” Advisor: Dr. Mufti Mahmud.

Rajshahi University of Engineering & Technology (RUET), Rajshahi, Bangladesh Mar 2014 – Nov 2018

B.Sc. in Computer Science & Engineering; CGPA: **3.70/4.0**

Thesis: “Cancer Diseases Prediction using Data Mining Techniques.”

PUBLICATIONS

Peer-Reviewed Journal Articles

- [J2] **Islam, T.**, Ali, M.S., Hossain, M.M., Shafiullah, M., & Alam, M.S. (2026). OBES-Net: An Exhaustively Optimized Explainable Ensemble Framework for Robust Obesity Risk Prediction. IEEE Access, 14, 73321–73342. (Published)
- [J1] Morshed, A., et al., **Islam, T.**, Hussain, M. (2025). Ultrasound-Based AI for COVID-19 Detection: A Comprehensive Review of Public and Private Lung Ultrasound Datasets and Studies. Multimedia Tools and Applications, 1–55. (Published)

Peer-Reviewed Conference Papers

- [C3] **Islam, T.**, & Mahmud, M. (2026). NeuroAPS-Net: Neuro-Anatomically Aware Point Cloud Representation for Efficient Alzheimer’s Disease Classification. In Proc. IEEE International Joint Conference on Neural Networks (IJCNN/WCCI 2026). (Accepted) [arXiv:2604.22883]
- [C2] **Islam, T.**, Alqudah, J., Alghamdi, H.S., Brown, D.J., & Mahmud, M. (2026). Sampling Matters: An Empirical Study of Point Cloud Sampling Strategies for Brain MRI-Based Alzheimer’s Disease Classification. In Proc. AI & NeuroTech Workshop, IEEE IJCNN/WCCI 2026. (Accepted)
- [C1] Alqudah, J., **Islam, T.**, Nafisa, N., Albadi, N., Brown, D.J., & Mahmud, M. (2026). Can We Trust Gaze? A Reliability and Fairness Study of Webcam Gaze Estimation in Online Learning. In Proc. IEEE/WIC/ACM International Conference on Web Intelligence and Intelligent Agent Technology (WI-IAT 2026). (Accepted)

Preprints & Manuscripts Under Review

- [P4] **Islam, T.**, Mahmud, M., & Brown, D.J. (2026). DAPGeN: A Disease-Aware Point Cloud Pipeline for Efficient Alzheimer’s Disease Classification in Resource-Constrained Clinical Settings. Submitted to IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2027), Applications Track. (Under Review)
- [P3] Ali, M.S., & **Islam, T.** (2025). Heart Failure Prediction: An Explainable Cross-Validated Comparison of Several Machine Learning Models. medRxiv. (Under review)
- [P2] **Islam, T.**, Asgar, S., & Mahmood, S. (2025). LungEEO: An Optimized Explainable Ensemble Framework for Lung Cancer Prediction. Preprints. (Under review).
- [P1] **Islam, T.**, Asgar, S., & Mahmood, S. (2025). Lung Cancer Prediction Using Machine Learning on Structured Clinical Data: A Systematic Review of Diagnosis, Risk, and Survival Models. Preprints (Under review)

Patent (Pending): Islam, T., Ali, M.S., & Shafiullah, M. “OBES-Net: An Exhaustively Optimized Explainable Ensemble Framework for Robust Obesity Risk Prediction.” KFUPM Ref. 2025-574 (Submitted).

RESEARCH EXPERIENCE

Graduate Researcher — Geometric Deep Learning for Neuroimaging

Aug 2024 – Present

Dept. of Information & Computer Science, KFUPM · Advisor: Dr. Mufti Mahmud

- Designed an end-to-end pipeline (Anatomical Priority Sampling) converting T1-weighted MRI into anatomically labelled 2D point clouds, producing the ADNI-2DPC dataset.
- Developed NeuroAPS-Net, a lightweight ROI-aware point cloud classifier for Alzheimer’s disease vs. cognitively normal status, achieving 84.85% accuracy at 1.48 ms inference and 234.6 MB peak GPU memory — a ~26× memory reduction over DGCNN at comparable/better accuracy.
- Conducted a systematic ablation of five point cloud sampling strategies (uniform, random, confidence-aware, reliability-aware, boundary-aware) across PointNet/PointNet++ and three point densities, showing sampling reliability outweighs point density for diagnostic accuracy (Sampling Matters).
- Translated the above research into a governance-aware clinical prototype (NeuroAPS Clinical Research Workspace) mapped to ITU-T Y.3172 and Saudi national AI/health policy (NSDAI, HSTP, PDPL) as part of Team NeuroCloud AI.

- Extended this line of work into DAPGeN, a 3D disease-aware point cloud pipeline using SynthSeg-derived population-level AD–CN effect maps (rather than heuristic 2D priors) to concentrate sampling on disease-relevant subregions; under a leakage-free, multi-seed protocol, DAPGeN reaches 90.3% balanced accuracy on ADNI, exceeds a matched 3D CNN baseline (87.4%) while using 3.1× fewer FLOPs and 3.5× less memory, and generalizes to an independent cohort (OASIS-1) (**submitted to WACV 2027**).

Independent Research Collaborations — Explainable & Trustworthy ML

2025 – 2026

- Led OBES-Net: exhaustive hyperparameter optimization (50 configurations, 9 algorithms) and systematic comparison of voting vs. stacking ensembles for obesity risk prediction, reaching 99.10% accuracy with integrated SHAP/LIME explainability; accepted in IEEE Access and filed as a KFUPM patent.
- Co-developed the Gaze Reliability Index (GRI), a no-reference audit metric quantifying webcam gaze-signal trustworthiness in online learning; showed 71% of signal-quality variance is explained by student identity alone and that unreliable signal biases human engagement labels, with implications for fairness in vision-based educational AI (accepted, WI-IAT 2026).
- Authored/Co-authored ensemble/explainable-AI studies on lung cancer risk (LungEEO), heart failure prediction, and a systematic review of ML for lung cancer diagnosis and survival modelling.

Undergraduate Researcher — Data Mining for Disease Prediction

2017 – 2018

Dept. of CSE, RUET · Supervisors: M.K. Paul; Dr. A.H.M.S. Sattar

- Built classification models for cancer risk prediction using structured clinical data as part of undergraduate thesis.

HONORS, AWARDS & GRANTS

- Full Graduate Scholarship, M.S. in Computer Science, Dept. of Information & Computer Science, KFUPM (Aug 2024 – Present)
- Travel Grant, IJCNN/WCCI 2026 (International Joint Conference on Neural Networks)
- Finalist, AI Readiness Hackathon — KSA (Team NeuroCloud AI), national submission mapping an AD-screening pipeline to ITU-T Y.3172 and Saudi AI/health policy
- Dean’s Scholarship, RUET — awarded 3 of 8 semesters for CGPA above 3.75/4.0
- University Technical Scholarship, RUET — for consistent academic performance

TEACHING EXPERIENCE

Instructor & Teaching Assistant, Dept. of Information & Computer Science, KFUPM Aug 2024 – Present

ICS 104 (Intro. Programming, Python/C – Lab, **Instructor**); ICS 504 Deep Learning, ICS 590 Special Topics in Deep Learning, COE 292 Intro to AI (TA)

Lecturer (Study Leave), Dept. of CSE, Northern University Bangladesh

Oct 2019 – Present

AI & Expert Systems, Algorithms, Image Processing & Computer Vision, Data Structures, C Programming

Part-time Lecturer, Dept. of TMDM, Bangladesh University of Textiles

May 2019 – Jul 2020

C Programming, Database Management Systems

Lecturer, Dept. of CSE, North Bengal Int’l University, Rajshahi, Bangladesh

Jan 2019 – Apr 2019

Java, Data Structures, Microprocessors

TECHNICAL SKILLS

Programming: Python, C/C++, Java, Assembly

ML/DL Frameworks: PyTorch, TensorFlow, Caffe, MONAI, Scikit-learn

Explainable AI: SHAP, LIME

Neuroimaging & Analysis: FSL (FAST/BET), SynthSeg, MATLAB, Weka, R, Google Colab, MySQL

PROFESSIONAL SERVICE & ACTIVITIES

- Reviewer / active contributor, peer collaborations across obesity, lung cancer, and cardiac ML risk-prediction studies (2025)
- Participated in Solution Hackathon held at KFUPM (2025)
- Participated in 3-Day Workshop on Introduction to the Inclusive Education Toolkit funded by the British Council’s Going Global Partnerships programme, 22-24 June, 2025
- Vice-President, Greater Khulna Association of RUET (GKA-RUET)
- Volunteer, National Collegiate Programming Contest (NCPC)